Valaciclovir for the Suppression of Recurrent Genital Herpes Simplex Virus Infection: A Large-Scale Dose Range–Finding Study

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A randomized, double-blind study of valaciclovir for suppression of recurrent genital herpes was conducted among 1479 immunocompetent patients. Patients were randomized to receive valaciclovir (250 mg, 500 mg, or 1 g once daily, or 250 mg twice daily), acyclovir (400 mg twice daily), or placebo, for 1 year. All valaciclovir dosages were significantly more effective than placebo at preventing or delaying recurrences ($P < .0001$). There was a dose-response relationship ($P < .0001$) across the once-daily valaciclovir regimens. Twice-daily valaciclovir and acyclovir were similar in effectiveness. Subgroup analysis showed that patients with a history of <10 recurrences per year were effectively managed with 500 mg of valaciclovir once daily. One gram of valaciclovir once daily, 250 mg of valaciclovir twice daily, or 400 mg of acyclovir twice daily were more effective in patients with ⩾10 recurrences per year. Safety profiles of all treatments were comparable. Thus, valaciclovir is highly effective and well tolerated for suppression of recurrent genital herpes. Once-daily regimens offer a useful option for patients who require suppressive therapy for management of genital herpes.

Genital herpes simplex virus (HSV) infection continues to be a growing public health problem, affecting millions of people worldwide and causing significant morbidity, particularly in sexually active young adults [1–3]. After initial infection, the virus establishes latency within the sensory ganglia; >90% of HSV-2–infected persons have clinical recurrences, and nearly all experience subclinical recurrences [4]. Suppressive therapy with acyclovir is an established and effective form of management for recurrent HSV infections, particularly for individuals with frequent or severe recurrences [5]. Studies of daily suppressive acyclovir therapy have now extended to 10 years and have confirmed that this form of management is effective and extremely well tolerated [6, 7].

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Informed consent was obtained from patients prior to enrollment into the study, in accordance with the human experimentation guidelines of the US Department of Health and Human Services and of relevant local authorities. Ethics committee approval was obtained at each study site before recruitment commenced.

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The acyclovir dosage used most commonly for suppression among immunocompetent adults is 400 mg twice daily. Although once-daily therapy with acyclovir (800 mg/day) is more effective than placebo [8], there have been no controlled trials comparing once-daily acyclovir regimens with more frequent acyclovir regimens. Moreover, one uncontrolled study suggested that once-daily acyclovir (800 mg) was suboptimal [9]. For many people, once-daily therapy would be a more convenient long-term approach for management of HSV infection.

Valaciclovir (Valtrex, Zelitrex; Glaxo Wellcome, Research Triangle Park, NC, and Glaxo Wellcome, Greenford, UK) is the L-valine ester of acyclovir. After oral administration and absorption, valaciclovir is rapidly and virtually completely converted to acyclovir [10]. The bioavailability of acyclovir after oral administration of valaciclovir in humans is 3- to 5-fold greater than can be achieved with oral acyclovir itself [10, 11]. This enhanced acyclovir bioavailability suggests that once-daily valaciclovir would have excellent efficacy among immunocompetent patients with genital herpes. In previous studies, a once-daily dosage of 500 mg of valaciclovir was shown to be effective in the prevention or suppression of recurrent episodes of genital herpes [12, 13].

We conducted a placebo-controlled study to investigate the dose-response relationship for a range of once-daily valaciclovir dosages given for ⩽1 year among immunocompetent persons with frequently recurring genital herpes. Twice-daily dosage regimens of acyclovir and valaciclovir were also evaluated.

Materials and Methods

Study population. Men and women aged ⩾18 years with a history of six or more genital HSV recurrences per year were

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enrolled in this multicenter, randomized, double-blind study to evaluate the efficacy and safety of oral valaciclovir, oral acyclovir, and placebo for the prevention and suppression of recurrent episodes of genital HSV infection. Those who had been receiving suppressive therapy with acyclovir must have stopped therapy and must have had an HSV recurrence ⩽3 months after stopping acyclovir. This recurrence also had to be ⩽3 months before enrollment into the study. Patients had to have a history of HSV infection involving genital, perianal, or closely related sites, and the history had to be documented by a positive HSV culture, direct antigen test, Tzanck smear, immunofluorescence assay, or written confirmation of genital HSV infection by a primary care physician. Participants were categorized at entry as having a history of <10 or ⩾10 recurrences per year when not receiving suppressive therapy. Patients were not eligible if they had significant hepatic or renal impairment, were pregnant or nursing mothers, were known to be immunocompromised, were hypersensitive to acyclovir or valaciclovir, or were receiving systemic or topical antiviral or immunomodulatory treatments or probenecid.

Drug administration. Patients were screened and randomized to 1 of 6 treatment arms (valaciclovir: 250 mg once daily, 500 mg once daily, 1 g once daily, or 250 mg twice daily; acyclovir: 400 mg twice daily; or placebo) in accordance with a 2:2:2:2:2:1 treatment allocation, with each active-treatment arm twice the size of the placebo arm. Study treatment started immediately, and double-blind therapy was given for 52 weeks. Clinically confirmed recurrences were treated with episodic valaciclovir for 5 days.

Study procedures. At enrollment, each participant was examined, and a directed medical history, including information on the frequency and duration of genital HSV infection episodes, was obtained. Patients were scheduled to come to the clinic at monthly intervals, with additional visits <24 h after the first signs of an HSV recurrence (day 1 of the recurrence). If lesions had reached the macular/papular stage of development, double-blind therapy was discontinued temporarily, and the patient was given episodic valaciclovir for 5 days. Patients returned to the clinic on day 5 of the recurrence for assessment of lesion healing and resolution of pain or other discomfort. Throughout the study, diaries were used by patients to record any recurrences of genital HSV infection, use of concomitant medications, and adverse experiences.

Hematology (hemoglobin concentration and white blood cell and platelet counts) and clinical chemistry (creatinine, alkaline phosphatase, and aspartate aminotransferase or alanine aminotransferase levels) parameters were evaluated at enrollment and at months 6 and 12; the serum creatinine level was also measured at months 3 and 9. Female patients were tested for pregnancy at enrollment, at 3-month intervals throughout the study, and whenever pregnancy was suspected.

Plasma acyclovir concentrations at steady state were determined from blood samples taken at 3, 6, 9, and 12 months. The times of sample collection, doses of study drug taken within the previous 24 h, and the patient's weight were recorded. Acyclovir concentrations were measured by RIA [14]. The concentration-time profiles were analyzed to estimate representative peak acyclovir concentration (C_{max}) and daily area under the plasma acyclovir–concentration time curve (area under the curve, AUC) for each treatment group.

Statistical analysis. All analyses were conducted by use of Statistical Analysis Systems software (SAS Institute, Cary, NC) and were performed by the clinical statistics department of Glaxo Wellcome. The primary efficacy end point was the time to first recurrence of genital HSV infection, defined as the number of days from randomization until the first onset of lesions. For patients who did not reach this end point, censored event-free times were calculated to be equal to the number of days until the last day on which no event was recorded.

The statistical sample-size estimation was based on the primary efficacy end point described above. If the hazard functions are assumed to be proportional, 200 patients per active-treatment arm provided 80% power to detect the difference between 60% and 45% in the proportions of patients who were recurrence-free at 1 year (360 days) in 2 active-treatment arms, with a 10% allowance for premature withdrawals [15]. The sample size of 100 placebo patients provided >80% power to detect the difference between 45% and 5% in the proportion of patients recurrence-free at 1 year in a comparison of an active-treatment arm with the placebo arm.

The distributions of time to first recurrence of genital HSV infection were estimated by use of the Kaplan-Meier product-limit method. The differences between specified treatment groups were estimated and tested by use of the Cox proportional hazards model, which adjusted for center and prior HSV recurrence frequency.

The type I error was controlled by limiting hypothesis testing to comparisons specified in advance and by performing these as a series of conditional tests. These prespecified comparisons were conducted in 3 steps. Conditional hypothesis testing within steps 1 and 3 was performed by use of a step-down procedure. Analyses in steps 2 and 3 were conditional on the results obtained from step 1. The comparisons conducted within the 3 steps are listed below.

In step 1, the efficacy of each valaciclovir dosage was established by testing these dosages against placebo in a preassigned order. No further hypothesis testing was to be conducted if a nonsignificant result was obtained in the sequence.

In step 2, the most effective dosage was selected from the valaciclovir once-daily dosages after investigation of the nature of any dose-response relationship. The hypothesis testing conducted in this step was conditional on efficacy being established in two or more of the once-daily dosages in step 1.

In step 3, the valaciclovir twice-daily dosage was compared with the most effective valaciclovir once-daily dosage and with acyclovir, in a preassigned order. The hypothesis testing in this step was conditional on a significant result being obtained in step 1.

An analysis of crude proportions of patients recurrence-free at 12 months was conducted to complement the survival analysis. Patients who withdrew from the study before 12 months with an unknown recurrence status were counted as having had a recurrence during the study in this analysis. The Cochran-Mantel-Haenszel test, stratified by center, was used to test for prespecified treatment differences in the crude proportions of patients recurrence-free at 12 months.

Since prior HSV recurrence frequency was known to be a strong prognostic factor affecting time to first HSV recurrence, subgroup analyses were planned to estimate the treatment differences within subgroups of patients with <10 and ⩾10 HSV recurrences per year.

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Table 1. Demographic characteristics and disease history of patients.

Variable	VACV250bid (n = 274)	VACV1000 (n = 269)	VACV500 (n = 266)	VACV250 (n = 269)	ACV400bid (n = 267)	PBO (n = 134)
Median age, years (range)	34 (18–85)	34 (21–71)	34 (17–74)	34 (18–81)	33 (20–73)	33 (19–58)
Sex (%)						
Male	48	49	46	53	47	41
Female	52	51	54	47	53	59
Median time from first episode HSV, years (range)	5.6 (0.3–38.2)	6.1 (0.3–42.2)	4.6 (0.3–49.3)	4.9 (0.1–42.2)	4.6 (0.3–42.2)	5.7 (0–32.5)
HSV recurrences per year when not receiving suppressive therapy (%)						
<10	53	55	65	58	51	51
$\geqslant 10$	47	45	35	42	49	49
Use of suppressive acyclovir in previous year (%)	15	15	15	13	17	16
Median time on suppressive therapy, years (range)	0.5 (0.1–4.4)	0.4 (0–12.2)	0.5 (0–3.4)	0.4 (0.1–2.4)	0.4 (0–11.4)	0.5 (0.2–4.5)

NOTE. VACV250bid, valaciclovir 250 mg twice daily; VACV1000, valaciclovir 1 g once daily; VACV500, valaciclovir 500 mg once daily; VACV250, valaciclovir 250 mg once daily; ACV400bid, acyclovir 400 mg twice daily; PBO, placebo.

Differences between treatments or changes over time in hematology or clinical chemistry values were assessed by means of quartile plots of each parameter from samples obtained at clinic visits. Hodges-Lehmann estimates and 95% confidence intervals were used to assess changes from baseline and any treatment differences at each scheduled visit.

All analyses except the subgroup analyses were based on the intent-to-treat (all patients randomized) population. All statistical tests were two-sided.

Results

Patient Characteristics

A total of 1479 patients were randomized to treatment at 53 study sites in Europe, the United States, and Australasia. Of these, 274 patients received 250 mg of valaciclovir twice daily, 269 received 1 g of valaciclovir once daily, 266 received 500 mg of valaciclovir once daily, 269 received 250 mg of valaciclovir once daily, 267 received 400 mg of acyclovir twice daily, and 134 received a placebo (table 1). Of the 1479 enrollees, 1050 (71%) completed the protocol. The most common reasons for premature discontinuation were loss to follow-up (9.2%), withdrawal of consent (8.5%), adverse experiences (5.1%), and protocol violation (2.3%). These reasons were similarly distributed across the 6 treatment groups.

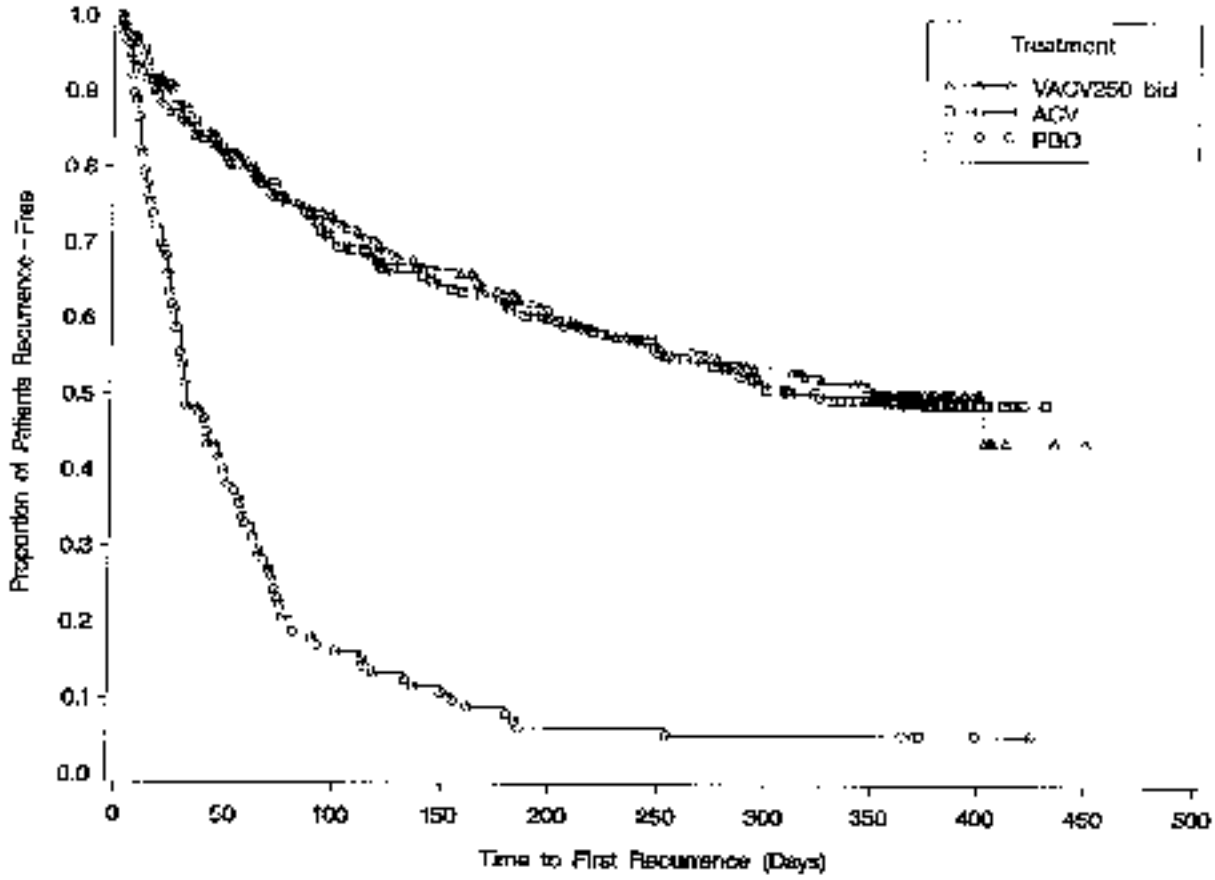
The demographics were similar across all 6 treatment groups. Overall, there were 770 women (52%) and 709 men (48%). The median time from the initial episode of HSV was 5.2 years, and 15% of patients had received suppressive acyclovir therapy during the previous 12 months, for a median time of 5 months. There was an imbalance across the treatment groups in prior HSV recurrence frequency: A lower proportion of patients (35%) experienced $\geqslant 10$ recurrences per year in the group who received 500 mg of valaciclovir once daily than in the other treatment groups (42%–49%).

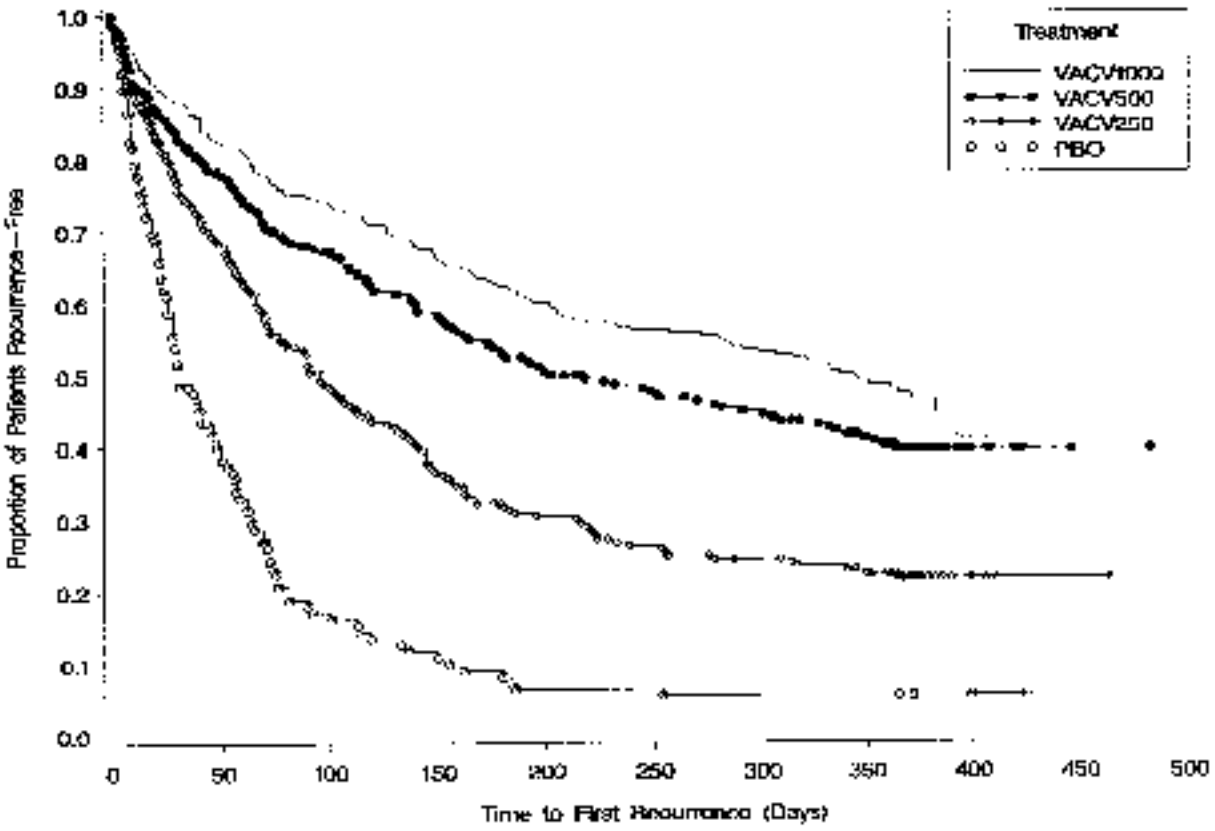
Efficacy

Compared with placebo, all dosages of valaciclovir and acyclovir were effective in suppressing HSV recurrences ($P <$ .01 for comparison to placebo for all treatment arms) (figure 1). The effects of twice-daily dosages of valaciclovir and acyclovir were indistinguishable from each other (figure 1A). The Kaplan-Meier estimates of proportions of patients recurrence-free at the end of 1 year were 50% for 250 mg of valaciclovir twice daily, 48% for 1 g of valaciclovir once daily, 49% for 400 mg of acyclovir twice daily, and 40% for 500 mg of valaciclovir once daily. Estimated proportions decreased to 22% for 250 mg of valaciclovir once daily and 5% for placebo (figure 1B).

The findings from the Kaplan-Meier estimates were confirmed by the intent-to-treat analysis of time to first HSV recurrence (table 2). Adjustments were made in the Cox proportional hazards model for the baseline imbalance in prior HSV recurrence frequency in this analysis. All valaciclovir-placebo treatment comparisons were highly significant ($P <$ .0001), indicating that each valaciclovir dosage was efficacious (step 1) and that each delayed or prevented HSV recurrences, compared with placebo. Moreover, decreasing hazard ratios indicated that, for once-daily valaciclovir treatment, efficacy increased with increasing dose. There was a dose-response relationship across the range of 3 valaciclovir once-daily groups ($P <$ .0001; step 2, table 2), confirming that increasing the valaciclovir once-daily dose enhanced efficacy. There was no significant difference between 1 g of valaciclovir once daily and

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Table 2. Cox's proportional hazards analysis of time to first HSV recurrence for the intent-to-treat group.

Treatment comparison	Hazard ratio (95% CI)	*P*
Step 1		
VACV250bid vs. PBO	0.21 (0.16–0.27)	<.0001
VACV1000 vs. PBO	0.22 (0.17–0.29)	<.0001
VACV500 vs. PBO	0.29 (0.22–0.37)	<.0001
VACV250 vs. PBO	0.46 (0.36–0.59)	<.0001
ACV400bid vs. PBO	0.21 (0.16–0.27)	—*
Step 2		
Dose-response hypothesis	—	<.0001
VACV1000 vs. VACV500	0.78 (0.61–0.99)	—*
Step 3		
VACV250bid vs. VACV1000	0.95 (0.74–1.22)	.68
VACV250bid vs. ACV400bid	1.01 (0.74–1.30)	—*
HSV recurrence history (covariate)		
<10 vs. ⩾10 recurrences/year	0.75 (0.65–0.87)	—*

NOTE. VACV250bid, valaciclovir 250 mg twice daily; VACV1000, valaciclovir 1 g once daily; VACV500, valaciclovir 500 mg once daily; VACV250, valaciclovir 250 mg once daily; ACV400bid, acyclovir 400 mg twice daily; PBO, placebo; CI, confidence interval.

* No hypothesis test was specified in the protocol, to avoid multiple significance testing and type 1 error.

250 mg twice daily ($P = .68$; step 3, table 2). A valaciclovir dosage of 250 mg twice daily or 1 g once daily and 400 mg of acyclovir twice daily were comparable in efficacy, reducing the recurrence rate by 78%–79%. A dosage of 500 mg of valaciclovir once daily reduced the recurrence rate by 71%. A valaciclovir dosage of 250 mg once daily reduced recurrences by 54% compared with placebo.

Cochran-Mantel-Haenszel analysis of patients recurrence-free at 12 months (based on crude proportions). These analyses were conducted because the percentage of patients lost to follow-up before 12 months without having had a recurrence was higher for patients allocated to active-treatment arms (17%–22%) than for those receiving placebo (10%). Since survival analysis assumes that the withdrawal rates across treatment groups are comparable, the active treatment–placebo comparisons are weakened. The analysis of crude proportions does not rely on this assumption and provides a worst-case analysis in comparisons with placebo. The crude percentages of patients known to be recurrence-free at the end of 1 year were 34% for those receiving 250 mg of valaciclovir twice daily, 34% for those receiving 1 g of valaciclovir once daily, 35% for those receiving 400 mg of acyclovir twice daily, 28% for those receiving 500 mg of valaciclovir once daily, 15% for those receiving 250 mg of valaciclovir once daily, and 5% for those receiving placebo. In the statistical analysis, all valaciclovir versus placebo comparisons were highly significant (250 mg of valaciclovir once daily, $P = .03$; for all other comparisons, $P < .001$). The effects of valaciclovir (250 mg twice daily, 1000 mg once daily, or 500 mg once daily) and 400 mg of acyclovir twice daily were similar.

Treatment effects in subgroups of patients with <10 and ⩾10 recurrences per year. Patients who had entered the study with <10 recurrences per year when not receiving suppressive therapy had a recurrence rate during the study that was 25% lower than the recurrence rate in patients who had entered the study with ⩾10 recurrences per year (table 2).

An analysis of treatment differences in the HSV recurrence frequency subgroups (<10 or ⩾10 recurrences per year) is shown in table 3. For all treatment dosages including twice-daily acyclovir and valaciclovir, suppression was less complete among those with a history of more frequently recurring disease. The differences between groups were especially marked in the groups receiving dosages of 500 mg and 250 mg once daily (figure 2).

Safety

The incidence and nature of adverse experiences reported were similar for each valaciclovir group and for the acyclovir and placebo groups (table 4). Most adverse experiences were mild. The most frequent adverse experiences reported in all groups were headache, rhinitis, infection, and flu-like syndrome. Sixty serious adverse experiences were reported by 48 patients during the study; distributions were similar across all treatment groups. The most commonly reported serious adverse experience was depression (8

Table 3. Cox's proportional hazards analysis of time to first HSV recurrence for HSV-recurrence–history subgroups.

	Hazard ratio (95% CI)	
Treatment comparison	<10 recurrences/year	⩾10 recurrences/year
VACV250bid vs. PBO	0.12 (0.08–0.18)	0.33 (0.23–0.47)
VACV1000 vs. PBO	0.16 (0.11–0.23)	0.27 (0.19–0.40)
VACV500 vs. PBO	0.18 (0.13–0.26)	0.45 (0.31–0.66)
VACV250 vs. PBO	0.32 (0.23–0.45)	0.63 (0.45–0.90)
ACV400bid vs. PBO	0.16 (0.11–0.23)	0.26 (0.18–0.38)
VACV1000 vs. VACV500	0.88 (0.64–1.21)	0.61 (0.42–0.88)

NOTE. VACV250bid, valaciclovir 250 mg twice daily; VACV1000, valaciclovir 1 g once daily; VACV500, valaciclovir 500 mg once daily; VACV250, valaciclovir 250 mg once daily; ACV400bid, acyclovir 400 mg twice daily; PBO, placebo; CI, confidence interval.

Figure 1. Kaplan-Meier plots of time to first HSV recurrence. **A**, VACV250bid, valaciclovir 250 mg twice daily; ACV, acyclovir 400 mg twice daily; PBO, placebo. **B**, VACV1000, valaciclovir 1 g once daily; VACV500, valaciclovir 500 mg once daily; VACV250, valaciclovir 250 mg once daily; PBO, placebo. $P < .01$ for each treatment arm vs. placebo.

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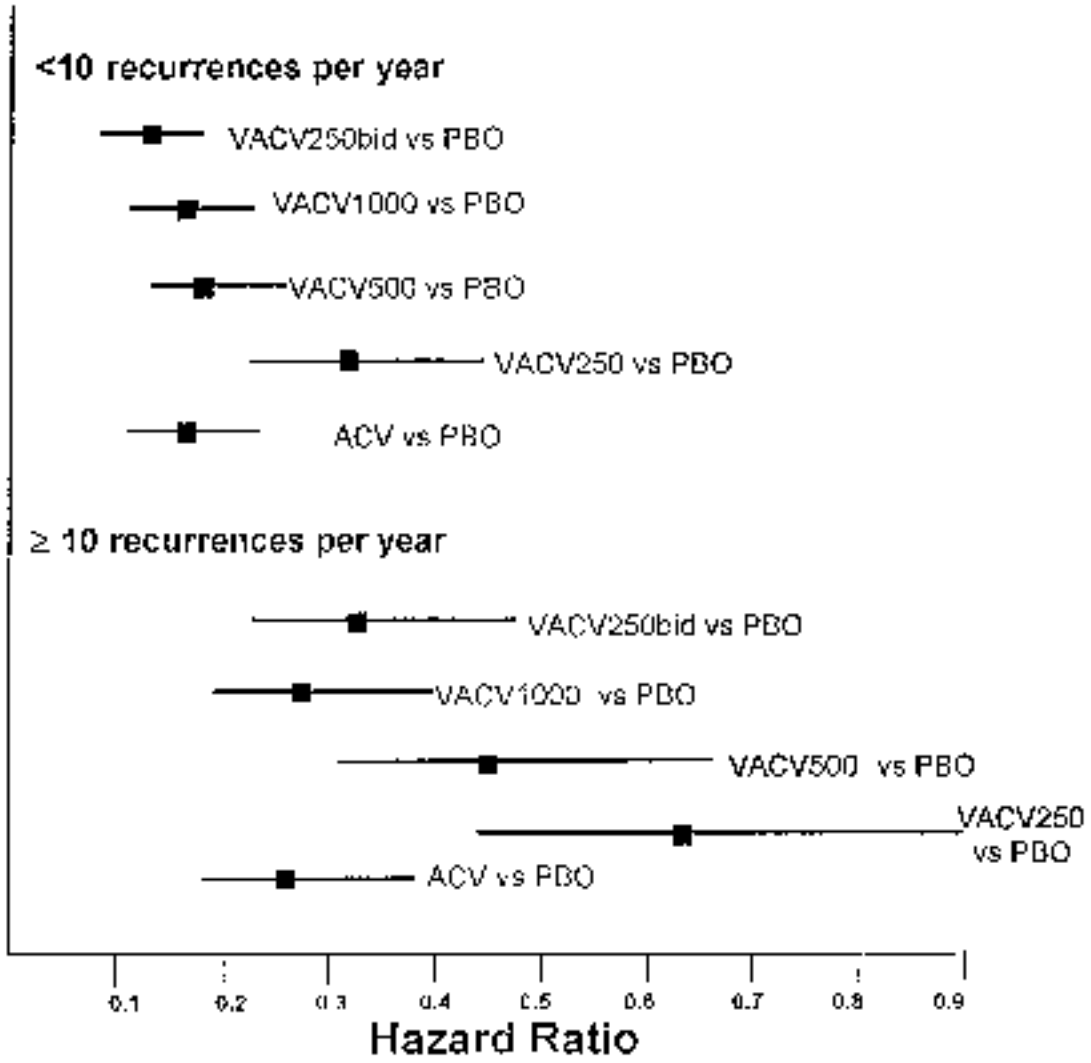


Figure 2. Hazard ratios and 95% confidence intervals for analysis of subgroups defined by no. of HSV recurrences per year (<10 and ⩾10) among patients not receiving suppressive therapy. VACV250bid, valaciclovir 250 mg twice daily; VACV1000, valaciclovir 1 g once daily; VACV500, valaciclovir 500 mg once daily; VACV250, valaciclovir 250 mg once daily; ACV, acyclovir 400 mg twice daily.

patients). Only 1 adverse experience, that of an acyclovir recipient, was considered possibly attributable to the study drug. One patient in the group receiving 250 mg of valaciclovir once daily died of bronchial asthma during the study. This death was not considered to be attributable to valaciclovir.

There were no clinically significant differences between treatment groups with regard to distribution or changes from screen for any clinical chemistry or hematology parameter. In particular, valaciclovir administration was not associated with changes in renal function.

Plasma Acyclovir Concentrations and Pharmacokinetics

The C_{max} values and daily AUCs determined from pharmacokinetic analyses of samples from 1297 patients were consistent with data found in prior phase I trials [10]. The C_{max} values were 1.57, 5.13, 2.48, 1.36, and 0.67 μg/mL, and the daily AUC values were 12.39. 18.13, 10.35, 4.95, and 7.60 h.μg/mL, for 250 mg of valaciclovir twice daily; 1 g, 500 mg, and 250 mg of valaciclovir once daily; and 400 mg of acyclovir twice daily, respectively. Peak acyclovir concentrations and AUCs increased with increasing once-daily valaciclovir dosages, a finding that was consistent with the efficacy dose-response relationship observed for the valaciclovir once-daily regimens.

Discussion

The study described here is the largest placebo-controlled investigation to date of any antiviral agent for the suppression

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Table 4. Adverse experiences, without regard to causality, reported by ⩾10% of patients in any treatment group.

Adverse experience	Patients reporting adverse experience (%)					
	VACV250bid	VACV1000	VACV500	VACV250	ACV400bid	PBO
Headache	35	35	38	35	37	34
Rhinitis	26	23	23	19	25	18
Infection	15	17	18	17	21	16
Flu-like syndrome	20	20	12	14	18	13
Pharyngitis	8	13	12	10	11	14
Nausea	9	11	11	12	12	8
Back pain	12	6	8	11	13	6
Diarrhea	8	12	9	7	12	14
Abdominal pain	9	11	9	6	7	6
Pain	6	9	11	9	8	4
Sinusitis	7	9	8	8	12	5
Accidental injury	7	9	7	7	10	3
Dyspepsia	6	10	3	6	7	8

NOTE. VACV250bid, valaciclovir 250 mg twice daily; VACV1000, valaciclovir 1 g once daily; VACV500, valaciclovir 500 mg once daily; VACV250, valaciclovir 250 mg once daily; ACV400bid, acyclovir 400 mg twice daily; PBO, placebo.

of recurrent genital HSV infection. It has demonstrated that valaciclovir is effective, even at low dosages, in reducing clinical reactivation of HSV infection. Perhaps of more importance, a dose-response relationship was seen between clinical reactivation of HSV and antiviral therapy. First, clinical effectiveness was better with the once-daily valaciclovir dosages of 1 g and 500 mg than with 250 mg once daily. Second, persons with ⩾10 recurrences per year had a lesser response to therapy at all dosages tested. For persons with ⩾10 recurrences per year prior to therapy, 1 g once daily was better than 500 mg once daily. These once-daily regimens were essentially equivalent to the twice-daily 250-mg valaciclovir and 400-mg acyclovir dosages.

The strengths of the study included its size and the similar demographics and baseline characteristics of the 6 treatment arms, which were well matched (except for HSV recurrence history). There were fewer patients in the valaciclovir 500 mg once-daily group who experienced ⩾10 recurrences per year when not receiving suppressive therapy. The Cox proportional hazards analysis adjusted for this imbalance.

Given the number of treatment arms, it was important to limit hypothesis testing, both to protect against type I error and to preserve the power of the study. Hypothesis testing was therefore performed on prespecified treatment comparisons in 3 steps, with a step-down procedure within steps 1 and 3. Thus, only specified comparisons were made. The first step was to establish efficacy of the valaciclovir treatment arms compared with placebo. Once efficacy was confirmed, analysis progressed to the second step—investigation of the nature of the once-daily dose-response relationship. The final step compared the once-daily dose chosen in step 2 with twice-daily valaciclovir and investigated the twice-daily arms of the study.

In this study, all dosages of valaciclovir investigated were highly efficacious compared with placebo, indicating that all of the regimens could be used for long-term suppression of HSV infection. However, the nature of the dose-response relationship (figure 2) clearly showed that a valaciclovir dosage of 250 mg once daily does not provide the degree of efficacy observed with the higher once-daily dosages. Only 22% of patients receiving the 250 mg once-daily dosage remained recurrence-free throughout the 12-month study period. Increasing the valaciclovir once-daily dosage to either 500 mg or 1 g provided a much greater degree of control over recurrences, with 40%–50% of patients being recurrence-free at 12 months. There was virtually nothing to distinguish the once-daily valaciclovir dosage of 1 g from either 250 mg of valaciclovir twice daily or 400 mg of acyclovir twice daily. The results of this study are comparable to those of previous studies of acyclovir (400 mg twice daily) in similar patient populations in which ~45% of patients were recurrence-free at 12 months [8, 16]. However, valaciclovir maintains this level of efficacy with a once-daily dosage regimen, probably as a result of its enhanced acyclovir bioavailability.

One of the important clinical observations in our study was that, while receiving therapy, patients with a history of <10 recurrences per year had a 25% lower recurrence rate than did those with a history of ⩾10 recurrences per year. Planned subgroup analyses were conducted to investigate whether the treatment effects and the dose response were consistent across recurrence-history subgroups (<10 or ⩾10 recurrences per

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year), as this was suspected in previous trials. The results of these analyses indicate that, for patients with <10 recurrences per year, in terms of clinical efficacy, there was little to differentiate the outcomes of a regimen of 500 mg of valaciclovir once daily versus 1 g once daily. Acceptable suppression of HSV recurrences was therefore achieved with a 500 mg once-daily dosage of valaciclovir in patients with less frequently recurring disease. However, patients with more frequent recurrences (≥10/year) were more effectively managed with a regimen of a higher once-daily dosage (1 g) of valaciclovir or with a twice-daily dosage of 250 mg of valaciclovir or 400 mg of acyclovir.

The safety profile of valaciclovir at all dosages was similar to those of acyclovir and placebo; reported adverse experiences were typically mild and infrequent. Although a dose-response relationship was observed in the efficacy of valaciclovir, there was no evidence of this in the adverse-experience profile. Indeed, this was true despite the fact that patients allocated to receive the highest valaciclovir dosages were in the study longer and were therefore subject to a longer period of safety monitoring. The nature of the adverse events was reminiscent of that in previous placebo-controlled studies, headache being the most commonly reported event. Laboratory safety monitoring also indicated that valaciclovir was well tolerated, with no clinically significant effects on hematologic or biochemical parameters.

In summary, this study, conducted with a large patient population, has demonstrated that a range of once-daily dosages of valaciclovir can significantly prevent and delay the recurrence of genital HSV infection over 1 year. For patients with a history of <10 recurrences per year, there was little to distinguish the clinical efficacy of once-daily valaciclovir dosages of 500 mg from that of 1 g, thus making 500 mg once daily an appropriate starting dosage. However, for patients with more frequently recurring disease (≥10 recurrences/year), a 1 g once-daily dosage of valaciclovir or more frequent doses of either 250 mg of valaciclovir or 400 mg of acyclovir would be preferable.

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